ZOO502: Subjective Midterm Spring, 2023.
Composed and compile by Farhan Ahmed

Name hormones that regulate water and ions? Marks3

- Antidiuretic hormone (ADH)
- Aldosterone
- Atrial natriuretic Hormone
- Calcitonin
- Parathormone

Adenohypophysis hormones name? Marks3

Two Types:

1-Hypothalamic releasing hormones (RHs)

Hypothalamic releasing hormones (RHs)
Corticotropin releasing hormone (CRH)
GH releasing hormone (GHRH)
TSH releasing hormone (TRH)

2-Neurohypophyseal hormones

Oxytocin
Antidiuretic hormone

What is synapse? Marks3 Synapses it's types Marks3

Synapse

The junctions between axon terminals of one neuron and the dendrites of another neuron are known as synapses. At these junctions, transmission of signals takes place from one neuron to another.

There are two **types of synapses**:

Electrical synapses

At electrical synapses, the plasma membranes of the pre- and postsynaptic cells are in close apposition. They are electrically coupled by particular proteins called gap junctions present within the membranes.

Chemical synapses

The chemical synapses are involved in chemical synaptic transmission involving neurotransmitters. This mode of synaptic transmission is more common as compared to the electrical synaptic transmission.

Myosin filaments? Marks5

- Myosin filaments are composed of myosin molecules, which are a type of motor protein that can bind to actin and use ATP to generate force and movement.
- Myosin filaments have a rod-shaped tail and a globular head that forms the cross-bridge with actin1. The cross-bridges are arranged in a helical pattern along the filament.

- Myosin filaments are found in the A band of the sarcomere, which is the basic unit of muscle contraction. They are anchored at the center by the M line, which contains proteins such as myomesin and creatine phosphokinase.
- Myosin filaments interact with actin filaments through a power stroke mechanism, in which the myosin head changes its conformation and pulls the actin filament towards the center of the sarcomere. This shortens the sarcomere and produces muscle contraction.
- Myosin filaments are classified into different types based on their structure and function. For example, myosin II is the main type found in skeletal and cardiac muscle, while myosin V is involved in intracellular transport.
- Myosin filaments are regulated by various factors, such as calcium, troponin, tropomyosin, phosphorylation, and ATP availability. These factors can modulate the binding affinity and activity of myosin with actin.
-

Range Fractionation? Marks5

Definition

In a sense organ, sensory receptors are arranged in an order of increasing sensitivity to different range of intensities of the stimulus. This hierarchical arrangement of receptors is known as range fractionation.

Range Fractionation and Stimulus Intensities

Each individual receptor, in a sense organ, covers only a fraction of the total dynamic range of the stimulus.

Receptors work together in an orderly way to provide discrimination of stimulus intensities.

Recruitment Phenomenon

One important implication of range fractionation is the recruitment phenomenon.

• The most sensitive receptors in the population produce a response at stimulus intensities that are just above the threshold. Above that intensity, the most sensitive receptors become saturated.

• If the stimulus energy is increased a little, the less sensitive receptors in the population will join in and respond.

• With still greater stimulus intensities, another, formerly inactive lower-sensitivity population

of receptors will join in.

• In this way, as the stimulus intensity is increased, receptors that are less and less sensitive will become active, until the least sensitive sensory fibers will finally be recruited, and all receptors will respond maximally.

• At that point, the system will be saturated and therefore unable to detect further increases in intensity.

Write a note on exons? Marks5

3. Axon

• An axon is a long and thick process that arises from the cell body and has a constant diameter.

• The lengths of the axons vary from few millimeters to more than a meter.

• The longest axons are seen in whales, where the axon of a single spinal motor neuron

may extend many meters from the base of the spine to the muscles in the tail fin.

Axon—Functions

• The axons are specialized to conduct signals away from the cell body. • They have evolved mechanisms that allow them to carry information for long distances with high fidelity and without loss.

Structural components of axons:

The axon contains a jelly-like semi-fluid substance called Axoplasm, surrounded by the plasma membrane which is called Axolemma.

Axon has mitochondria and ER but lacks Nissl's granules. So exons are not involved in protein synthesis.

Axon terminals:

• Each axon may divide into numerous branches at its termination. These branches of axon are called axon terminals.

• Each axon terminal has small extensions called telodendria with enlarged ends called terminal knobs. These knobs have granules or vesicles that contain neurotransmitters.

• The axon terminals allow transmission of signals to many other neurons, glands or muscle fibers simultaneously.

Note on exocrine glands? Marks5

Exocrine glands

Exocrine glands produce fluid secretions that are delivered through ducts onto the epithelial surfaces of the body.

Exocrine glands are more easily identified than endocrine glands because of their duct leading to the body surface.

The fluid secretions may be either proteins (enzymes) or mucous or both.

Examples:

• Salivary glands produce saliva that is delivered to the oral cavity for partial digestion of food through parotid and submandibular ducts.

• Pancreas produces enzyme-containing pancreatic juice that is delivered to the small intestine through pancreatic duct.

• Lacrimal glands produce tears that is delivered through lacrimal duct on the surface of eye to provide lubrication.

• Mammary glands produce milk that is delivered through lactiferous ducts to the nipples for nourishing the young.

Sarcomere? Marks5 Structure of sarcomere Marks5

Sarcomere

The regular arrangement of the thick and thin filaments creates a pattern of repeating light and dark bands.

This pattern gives a striped appearance to the muscle cell.

Each repeating unit is called a sarcomere and is the basic functional contractile unit of the muscle.

Structure of Sarcomere

• Each dark band in a sarcomere is called A band. This band is anisotropic i.e. it polarizes visible light.
• Each A band has a lighter stripe in its center which is called H-zone.
• The H-zone is bisected by a dark line called M-line.
• The light band is called I band. It is isotropic i.e. non-polarizing.
• The I band has a mid-line called Z-line.

A sarcomere is the region of a myofibril between two successive Z-lines.

Hormones of placenta? Marks3

Chorionic gonadotropin	Increases progesterone synthesis by corpus luteum
Placental lactogen	Stimulates fetal growth and development Increases mammary gland development in mother

Cells of thyroid gland? Marks3

It has two types of glandular cells:
Follicular cells which secrete two hormones:
✓ Triiodothyronine (T3)
✓ Tetraiodothyronine (T4) also known as thyroxine
Parafollicular cells which secrete one hormone:
o Calcitonin

Effects of neuromodulation? Marks5

The phenomenon of neuromodulation plays critical roles in the development of complex behavioral patterns e.g. happiness, revenge, reward, greed and exploration. Neuromodulation is
also involved in the processes of thinking, cognition, planning, learning and memory.
Many behavioral problems are also based on neuromodulation e.g. mood swings, sleep disturbances, feelings of stress, anxiety, depression and anger.

Sliding Filament Theory? Marks5

H. E. Huxley and A. F. Huxley proposed the sliding filament theory of muscle contraction in 1954.
This theory states that during muscle contraction the thin and thick filaments in sarcomeres slide and undergo shifting.
When a muscle contracts, the thin actin filaments actively slide along between the thick myosin filaments and move closer to the center of the sarcomere. As a result, the sarcomere becomes shorter.
When a muscle relaxes or is stretched, the overlap between thin and thick filaments is reduced, and the sarcomere elongates.

The changes in sarcomere length during contraction and stretch of a muscle, correspond to changes in muscle length.

Types of cilia? Marks5

Cilia are organelles found in eukaryotic cells that can be of two types: motile cilia and non-motile cilia. The overall basic structure of both types of cilia is the same, except the axoneme.

Non-motile:

Non-motile cilia are also known as primary cilia or sensory cilia. They do not have the ability to move, and mainly function as cellular antennas to regulate signaling pathways and maintain cell homeostasis. Most vertebrate cell types possess a single non-motile primary cilium. **Examples** of non-motile cilia are found on the tubular epithelial cells of the kidney and photoreceptors of the retina.

Motile cilia:

Motile cilia are cellular structures that move in a wave-like motion, aiding in fluid movement. They are found on the surface of various cells, including respiratory and reproductive tract epithelial cells, as well as protozoans like paramecium. Motile cilia consist of a central pair of single microtubules surrounded by nine pairs of double microtubules, forming a 9+2 axoneme. They play important roles in functions such as mucociliary clearance, cerebrospinal fluid circulation, egg transport, and establishing left-right asymmetry.

Brief note on axons

An axon is a long and thick process that arises from the cell body and has a constant diameter. The lengths of the axons vary from few millimeters to more than a meter.
The longest axons are seen in whales, where the axon of a single spinal motor neuron may extend many meters from the base of the spine to the muscles in the tail fin.

Axon—Functions

The axons are specialized to conduct signals away from the cell body.
They have evolved mechanisms that allow them to carry information for long distances with high fidelity and without loss.

Hair cilia marks3

Hair cilia are not technically known as cilia, but rather as a special type of non-motile cilia. They are small hair-like structures present on the surface of some eukaryotic cells. They are mainly involved in mechanoreception. Hair cells have a non-motile cilium called a kinocilium, which is involved in hearing and balance. Hair cilia are composed of microtubules and can function in locomotion, fluid movement, and sensory reception.

Fibers of Zonula and ciliary muscles? Or role of ciliary and fiber of zonula of eye marks3

FIBERS OF ZONULA

The fibers of zonula are tiny thread-like fibers that hold the eye's lens firmly in place. They are also known as zonular fibers or suspensory ligaments of lens. They are in contact with the ciliary muscle and help to support the lens and keep it in place.

Function/ Role

• The shape of the lens is changed by the fibers of the zonula that held the lens in place.
• These fibers can exert an outwardly directed tension on the perimeter of the lens.

CILIARY MUSCLES

The ciliary muscles are circular bands of smooth muscle that control the tightness of the fibers of zonula. They are located in the ciliary body, which is a ring-shaped structure that encircles the lens.

Function/ Role

- Ciliary muscles are attached with the fibers of zonula.
- Their contraction and relaxation adjust the amount of tension exerted on the lens.
- When the ciliary muscles contract, the lens is flattened by elastic tension exerted by the
  fibers of the zonula, which pull the perimeter of the lens outward.
- In this state, objects far
  from the eye are focused on the retina, but objects close by would be fuzzy.
- When the ciliary muscles relax, lens becomes more rounded. This focuses the objects close to the eye.

Functions of estrogen in pregnancy? Marks5

During pregnancy, estrogens exert mainly a proliferative function on most reproductive and associated organs of the mother. They cause:
(1) enlargement of mother's uterus
(2) enlargement of mother's breasts and growth of breast ductal structure
(3) enlargement of mother's external genitalia.
They also relax the pelvic ligaments of the mother, so that the pubic symphysis becomes elastic.
These changes allow easier passage of the fetus through the birth canal

Negative feedback with examples? Marks5

- The secretory activities of most endocrine tissues, especially those involved in maintaining homeostasis, are modulated by negative feedback.
- In this type of feedback, the hormone itself or products resulting from its action tend to suppress its further release.
- Negative feedback mechanisms ensure a proper level of hormone activity at the target tissue and prevent its over-activity by over-secretion.

Examples

Some examples of negative feedback in the endocrine system are:

- The regulation of thyroid hormones by the hypothalamus, the anterior pituitary gland, and the thyroid gland. When the thyroid hormones are high, they inhibit the production of TRH and TSH, which reduces their own secretion3.

- The regulation of blood calcium level by the parathyroid hormone. When the blood calcium level is high, it inhibits the secretion of parathyroid hormone, which lowers the blood calcium level45.
- The regulation of blood sugar level by insulin. When the blood sugar level is high, insulin is released by the pancreas, which facilitates the uptake of glucose by the cells and lowers the blood sugar level16.

Inflow of sodium ion (depolarization)

During an action potential, there is a rapid influx of sodium ions into the neuron, causing depolarization. This depolarization allows the electrical signal to propagate along the axon. The other options listed are not accurate descriptions of the changes that occur during an action potential.

function of myelin sheath? marks 3

- **Electrical insulation**: It prevents the leakage of electrical signals from the axon and increases the resistance of the membrane.
- **Increase the speed of conduction**: It reduces the capacitance of the membrane and allows the electrical signals to travel faster along the axon.
- **Space efficiency**: It reduces the diameter of the axon and allows more nerve fibers to be packed in a given space.
- **Saltatory conduction**: It enables the electrical signals to jump from one node of Ranvier to another, skipping the myelinated segments of the axon.

Corpus luteum role in pregnancy? marks 3

If the released ovum is fertilized and becomes implanted in the endometrium, the developing placenta begins to produce chorionic gonadotropin hormone that maintains an active corpus luteum, so that estrogen and progesterone secretion continues until the placenta fully takes over
the production of these hormones, at which time the corpus luteum degenerates.
In many other mammals, such as the rat, the corpus luteum, stimulated by prolactin, continues to grow and secrete estrogen and progesterone throughout the gestation period.

myofilaments type? marks3

Thick Filaments

• Thick filaments extend the entire length of A band.
• These filaments are about 16 nm in diameter and are composed of about 300 myosin molecules.

Thin Filaments

• The thin filaments are 7-8 nm thick and extend across the I band.
• They are composed chiefly of actin molecules.
• Thin filaments also overlap myosin filaments in the peripheral darker regions of the A band.

Name and explain hair cell cilia types? Marks5

Hair cells are named for the many cilia that project from the apical end of each cell. These cilia fall into two classes: **kinocilium and stereocilia**.

kinocilium

Each hair cell typically has a single kinocilium. The kinocilium has a "9 + 2" arrangement of internal microtubules that are similar to that of other motile cilia.

stereocilia

The stereocilia are structurally and developmentally distinct from the kinocilium.

- They are not formed of microtubules.
- They are formed of fine longitudinal actin filaments.
- Each hair cell has 20-300 nonmotile stereocilia.
- The stereocilia of a hair cell are arranged in order of increasing length from one side of the cell to the other.

difference between temporal and spatial summation? Marks5

Temporal summation

When two EPSPs occur at a single synapse in such rapid succession that the postsynaptic neuron's membrane potential has not returned to the resting potential before the arrival of the second EPSP, the EPSPs are added together. This is known as temporal summation.

Explanation

When a presynaptic terminal fire, the released transmitter produces an EPSP in the postsynaptic neuron by opening membrane channels for almost a millisecond. But the changed postsynaptic potential lasts up to 15 milliseconds. So, during rapid firing rate by the presynaptic terminal, the channels are opened again adding to the postsynaptic potential that increases to a greater level.

Spatial Summation

EPSPs produced nearly simultaneously by different synapses on the same postsynaptic neuron can also add together. This effect is known as spatial summation.

Explanation

Excitation of a single presynaptic terminal on the surface of a neuron almost never excites the neuron. The reason for this is that the transmitter substance released by a single terminal can cause an EPSP of about 0.5 to 1 millivolt. However, an EPSP of 10 to 20 millivolts is normally required to reach threshold for excitation.

structure of sarcoplasmic reticulum? Marks5

Structure of SR

SR is composed of two major parts:

(1) Terminal cisternae which are larger chambers. T-tubules are closely associated with terminal cisternae (with a separating distance of about 12 nanometers). These are the primary site of calcium release as they have voltage sensitive channel proteins that open

when action potential in T tubules activates them.
(2) Long longitudinal tubules that run between the terminal cisternae and surround the myofibrils. These are the locations where ion channels necessary for calcium ion bsorption (calcium pumps) are most abundant

phases of action potential Marks3

The course of the action potential can be divided into five phases:

➢ Stimulation

➢ Rising phase (Depolarization)

➢ Peak phase

➢ Falling phase (Repolarization)

➢ Undershoot (afterhyperpolarization)

abnormalities due to growth hormone Marks 3

igantism: excessive size and stature caused by hypersecretion of growth hormone during childhood (before puberty).
Acromegaly: enlargement of the bones of the head and of the extremities caused by hypersecretion of growth hormone beginning after maturity.
Dwarfism: abnormal underdevelopment of the body caused by insufficient secretion of growth
hormone during childhood and adolescence.

what is myofilaments Marks 3

Each myofibril is composed of myofilaments. Myofilaments are of two types, thin filaments and
thick filaments.
Thin filaments are composed of actin while the thick filaments are composed of myosin molecules.

difference between stigma and ocelli Marks5

between stigma and ocelli or Photoreceptive Structures

The complexity and arrangement of photoreceptors within various animals vary.
Eyespots or Stigma
The simplest photoreceptive structures are the Eyespots or Stigma found in some protozoa e.g.
euglena. Stigma is a bright red colored structure having carotenoid pigments. It gives a sense of
light and dark and helps in phototaxis.
eyecups or Ocelli
Cnidarians and flatworms have multicellular photoreceptive structures call eyecups or Ocelli. They consist of a cuplike depression containing photoreceptor cells. They cannot form image and only provide the animal a sense of direction.

hodgkins cycle 3 marks

Hodgkin cycle is a positive feedback loop in neurons that generates an action potential. It occurs when a membrane depolarization activates voltage-gated Na+ channels, which allow Na+ inflow and further depolarization. This cycle leads to a rapid rise in membrane potential to near VNa. The cycle is broken when voltage-gated K+ channels open and repolarize the membrane.

Factors affecting vascular resistance SVR 3marks

Vascular resisance is determined mainly by the muscle tone in the smooth muscle tissue of the tunica
media and the elasticity of the elastic fibers there.
Vasoconstriction (i.e., decrease in blood vessel diameter) increases SVR, whereas vasodilation (increase
in diameter) decreases SVR.

Structure of sarcomeres 5marks

Each dark band in a sarcomere is called A band. This band is anisotropic i.e. it polarizes visible light.

- Each A band has a lighter stripe in its center which is called H-zone.
- The H-zone is bisected by a dark line called M-line.
- The light band is called I band. It is isotropic i.e. non-polarizing.
- The I band has a mid-line called Z-line.

A sarcomere is the region of a myofibril between two successive Z-lines.

Phase response and tonic response 5marks

When neurons are stimulated continuously by a current of constant intensity, some neurons adapt rapidly and stop generating more APs after one or two initial action potentials at the beginning of the stimulus period.

- Two types:
- Phasic response
- Tonic response

Phasic response

The type of adaptation in which neurons adapt rapidly and stop generating more APs after initial one or two APs.

Tonic Response

The type of adaptation in which neurons adapt slowly and fire repetitively to a subthreshold stimulus, but with gradually decreasing frequency. These neurons are said to have a phasic response. Some neurons accommodate more slowly and therefore fire repetitively, but gradually decreasing frequency, in response to a prolonged constant-current stimulus. These neurons are said to have a tonic response.

□Long Question□

Structure of myosin and its function in muscle contraction marks5

Force is Produced by Rotation of Myosin Head

During muscle contraction, myosin heads make cross-bridges with actin filaments that pull the
thin filament toward the center of sarcomere. The force for pulling is produced by the partial rotation of myosin heads.

How Rotation is Produced?

The rotation is produced when the four sites M1 to M4 of myosin head interact sequentially with
the binding sites of actin filament.

Energy Storage in the Link

As the myosin head rotates against the actin
filament, the link is stretched elastically and stores
mechanical energy in the link due to tension
developed in it.

Transmission of Force to Thick Filament

The tension thus produced is transmitted to the thick filament through the neck of the myosin molecule. This tension provides force to shorten the sarcomere.

Detachment of Myosin Head

• When rotation of the head is complete, it dissociates from the actin filament and
rotates back to its relaxed position.
• Dissociation occurs when Mg2+ & ATP bind to head.
• ATP is then hydrolyzed, and myosin head changes conformation.
• Head rebind to a little farther site on the actin filament.

difference between concentric contraction and eccentric contraction marks5

Types of Isotonic Contractions

Based on how the length changes, isotonic contractions are further classified into two types:
1. Concentric contractions
2. Eccentric contractions

Concentric Contractions

If the muscle generates tension and the entire muscle shortens then it is a concentric contraction.
An example would be lifting a weight from your waist to your shoulder; the bicep muscle used
for this motion would undergo a concentric contraction.

Eccentric Contractions

If the muscle generates tension and the entire muscle lengthens, it is a eccentric contraction.
For example, lowering the weight from the shoulder to the waist, the bicep muscles would generate force but the muscle would be lengthening.
Eccentric contractions work to decelerate the movement at the joint.
Eccentric contractions can generate more force than concentric contractions.

write names of interceptor marks5

Photoreceptors: They detect light and are found in the eyes and some other organs.
Chemoreceptors: They detect chemicals and are found in the nose, tongue, and blood vessels.
Mechanoreceptors: They detect mechanical stimuli such as touch, pressure, vibration, and stretch. They are found in the skin, muscles, joints, and inner ear.
Thermoreceptors: They detect temperature changes and are found in the skin and hypothalamus.
Nociceptors: They detect pain and are found in most tissues of the body.
Proprioceptors: They detect body position and movement and are found in the muscles, tendons, and joints.
Osmoreceptors: They detect osmotic pressure and are found in the hypothalamus and kidneys.
Baroreceptors: They detect blood pressure and are found in the walls of the heart and blood vessels.

difference between osmoconformers and regulators marks3

Osmoregulators

Animals whose body fluid concentrations differ noticeably from the outside environment are called osomoregulators. They actively regulate water and salt to maintain an internal osmolarity different from the medium in which they are living. They discharge excess water in hypotonic environment and excrete salts in hypertonic conditions.

osmoregulators

Animals living in different environments e.g. marine, freshwater and terrestrial environments have distinct adaptations to regulate osmotic balance. Most vertebrates are osmoregulators. Only exceptions are elasmobranchs and hagfishes. Marine vertebrates keep their blood solute concentration hyposmotic to seawater while fresh water fishes maintain hyperosmotic composition as compared to freshwater.

what is role of t tubule marks3

The T tubule system provides the anatomic link between the surface membrane and the myofibrils deep inside the muscle fiber. When an action potential spreads over a muscle fiber membrane, a potential change also spreads along the T tubules to the deep interior of the muscle fiber.

Pace maker and types

Pacemaker

The pacemaker is the excitatory region which generates the rhythmical impulses that control the
rhythmicity of cardiac chambers.

Pacemaker Types

Two basic types of pacemakers in animals with pumping hearts:
• Neurogenic pacemakers
• Myogenic Pacemakers

Nonhuman Modalities name?

Many nonhuman animals also possess other sensory modalities that are unavailable to human beings.

Examples:

• **Distance thermoreception:** Pit organs are found in some snakes (e.g. pit vipers and rattle snakes) and can detect heat energy (infrared radiation) emitted from mammalian bodies. They use this sense to identify their prey due to temperature difference.

• **Electroreception** Detection of very low frequency electric signals. This sense is found in some electric fishes which use it to communicate with one another in murky waters.

• **Magnoreception:** Some animals appear to sense Earth's magnetic field with magnoreceptors. Magnoreceptors are found in many migratory birds which use it as a navigational guide..

Components of Excitatory and Conductive System ? marks5

The excitatory system of the heart comprises of the pacemaker region (also called sinoatrial or
S-A node), in which the rhythmical impulses are generated.

The conductive system includes:

• The atrial internodal pathways that conduct impulse from the S-A node to the atrioventricular (A-V) node.

• The A-V node, in which the impulse from the atria is delayed before passing into the ventricles.

• The A-V bundle, which conducts the impulse from the atria into the ventricles.

• The left and right bundle branches of A-V bundle.

• Purkinje fibers, which branch off from bundle branches and conduct the cardiac impulse to all parts of the ventricles.

Membrane physiology marks3

Membrane physiology is the study of the structure and function of the cell membrane, which is the barrier that separates the inside of the cell from the outside environment.

Photoreceptors marks3

Photoreceptors possess light-sensitive pigments which are carotenoids retinal and 3-dehydroretinal associated with opsin proteins to form rhodopsins. These pigments absorb photons of light energy and then produce a generator potential.

External energy, such as light, may strike any part of the body; but only the eyes contain sensory cells that can receive and respond to the stimulus of light and transduce photons into neuronal energy. Photoreceptors can be activated by a single photon.

Explain T4 cells marks3

Tetraiodothyronine (T4) cells are not a type of cells, but a type of hormone produced by the **thyroid gland**. T4 is also known as **thyroxine** or **tetraiodothyronine**, and it is one of the two main thyroid hormones that regulate the metabolism of all cells and tissues in the body. T4 is synthesized from **thyroglobulin**, a protein that is stored in the thyroid follicles, by the action of **thyroid peroxidase**, an enzyme that adds iodine to tyrosine residues.

Which mechanism involves in homeostasis. Give example marks3

The secretory activities of most endocrine tissues, especially those involved in maintaining homeostasis, are modulated by negative feedback.
In this type of feedback, the hormone itself or products resulting from its action tend to suppress
its further release.

Example:

1. Homeostatic regulation of water
2. Homeostatic regulation of temperature
3. Regulation of pH
4. Regulation of glucose concentration
5. Regulation of osmotic pressure
6. Regulation of oxygen level
7. Regulation of ion conc.

Taste buds and taste receptor. Marks5

Taste buds

• The gustatory organs of vertebrates are called taste buds.
• The taste bud is composed of about 50 modified epithelial cells including supporting cells (sustentacular cells), basal cells and taste receptor cells.
• The basal cells are progenitor cells that give rise to new taste receptors. They regularly generate new sensory taste receptor cells which
have an active life of only 10 days.

Taste Receptor Cells

• The outer tips of the taste cells are arranged around a minute taste pore.
• From the tip of each taste cell, several microvilli, or taste hairs, protrude outward into the taste pore to approach the cavity of the mouth.
• These microvilli provide the receptor surface for taste.
• Interwoven around the bodies of the taste cells is a branching terminal network of taste nerve fibers that are stimulated by the taste receptor cells.

non-soluble lipids or Lipid-insoluble hormone group Marks3

Lipid-insoluble hormones are hormones that cannot easily pass through the plasma membrane of cells because they are not soluble in lipids. They are usually derived from amino acids or peptides and have a chemical name ending in “-ine”. Some examples of lipid-insoluble hormones are insulin and glucagon, which regulate blood glucose levels,

epinephrine and norepinephrine, which trigger the fight-or-flight response, and thyroxine, which controls metabolism.

acetylcholine receptor

• Most of the knowledge about fast channels has been obtained through the studies of acetylcholine receptors and channels at the vertebrate neuromuscular junction.
• Acetylcholine receptors are of two types: Nicotinic AChRs and Muscarinic AChRs.
• Both receptors consist of two different types of channel proteins that are activated by acetylcholine. They are differentiated and named on the basis of their activation by muscarine and nicotine

Sensory transduction function? Marks3

Sensory receptor cells convert the physical or chemical energy of stimulus into electrical signals
i.e nerve impulse. This conversion is called sensory transduction.
All sensory transduction systems operate through similar cellular mechanisms and contain related molecules to transduce signal:
Functions:
• **Detection**: Sensing stimuli
• **Amplification**: Increasing signal strength
• **Encoding**: Converting signal into information
• **Transmission**: Sending signal to target

conductive Components of heart? Marks5

Components of Excitatory and Conductive System
The excitatory system of the heart comprises of the pacemaker region (also called sinoatrial or
S-A node), in which the rhythmical impulses are generated.
Components
Atrial internodal pathways: The atrial internodal pathways that conduct impulse from the S-A node to the atrioventricular (A-V) node.
A-V node: The A-V node, in which the impulse from the atria is delayed before passing into the ventricles.
A-V bundle: The A-V bundle, which conducts the impulse from the atria into the ventricles.
left and right bundle branches: The left and right bundle branches of A-V bundle.
Purkinje fibers: Purkinje fibers, which branch off from bundle branches and conduct the cardiac impulse to all parts of the ventricles.

LH hormone Marks3

Luteinizing hormone (LH)

Luteinizing hormone (LH) is a hormone that is produced by the **pituitary gland** in the brain and has important roles in both male and female reproductive systems. LH is regulated by another hormone called **gonadotropin-releasing hormone (GnRH)**, which is released by the **hypothalamus**.
Stimulates androgen production by gonads; ovulation, secretion of progesterone.

Functions of LH in males & female 3

LH is a hormone that stimulates the reproductive organs in both males and females. In females, LH triggers ovulation and progesterone production by the ovaries. In males, LH stimulates testosterone production and sperm formation by the testes. LH levels vary depending on age, sex, and menstrual cycle phase.

hormone of thyroid glands Marks3

Thyroxine: Maintain metabolic rate and oxygen.
Triiodothyronine: consumption; Growth and development.

Explanation

Thyroxine (T4) and triiodothyronine (T3) are hormones that are produced and released by the thyroid gland, a butterfly-shaped gland located in the front of the neck. They are both derived from the amino acid tyrosine and contain iodine atoms. Thyroxine has four iodine atoms, while triiodothyronine has three, hence their names.
Thyroxine and triiodothyronine are collectively known as thyroid hormones because they regulate the body's metabolism, growth, development, and temperature.

Hormone involved in regulation of water & ions 3

• These hormones include:
• Antidiuretic hormone (ADH)
• Aldosterone
• Atrial natriuretic Hormone
• Calcitonin
• Parathormone

Regulation of Water and Ions

• The organs of vertebrates which regulate water and ions are: kidneys, skin, gills, intestine and bone.
• The absorption or excretion of water and ions is regulated through these organs by the coordinated activity of many hormones mentioned above.

Sensation marks3

• Sensations are the subjective phenomena closely associated with the perception of stimulus.
• Sensations arise when sensory receptor cells, receiving a stimulus, transmit signals through the nervous system to particular parts of the brain that interprets or perceives these signals. The subjective description of these neuronal perceptions is termed as sensation.
• Interpretation of sensations is based on experience and learning i.e. previous exposure and its interpretation stored in the memory.

Examples

Examples of sensations are the feelings of pain, color, taste (e.g. sweet) noise, melody, bad or good odor etc.
Myosin structure & function 5

Hormones released by adrenal gland 3

Hormones released by Adrenal (Cortex) are….

Aldosterone

Increases blood Na+ levels; increase K+ secretion

Cortisol, corticosterone, cortisone

Role in carbohydrate metabolism

increase blood glucose levels

anti-inflammatory effects

Role of calcium in excitation and relaxation of action potential 3

Calcium plays an important role in both excitation and relaxation of action potential in neurons and muscle cells. Calcium ions enter the cell through voltage-gated calcium channels when the membrane depolarizes during an action potential. Calcium influx triggers the release of neurotransmitters from synaptic vesicles in the presynaptic neuron, which bind to receptors on the postsynaptic cell and modulate its activity. Calcium also causes the release of acetylcholine from motor neurons, which stimulates muscle contraction. On the other hand, calcium is removed from the cytoplasm by various mechanisms, such as calcium pumps and calcium-binding proteins, to restore the resting membrane potential and terminate the action potential. Calcium also activates potassium channels that help repolarize the membrane and relax the muscle. Thus, calcium is essential for both excitation and relaxation of action potential.

Components of homeostasis and one example?

Homeostasis is the process of maintaining a stable and relatively constant internal environment in living systems, despite changes in external or internal conditions. Homeostasis involves three components- the **receptor**, the **control center**, and the **effector**.

One example of homeostasis is the regulation of body temperature in mammals. The receptors are thermoreceptors in the skin and hypothalamus that detect changes in temperature. The control center is the hypothalamus, which compares the actual temperature with the set point (around 37°C) and sends signals to effectors. The effectors are sweat glands, blood vessels, and muscles that can alter heat production and heat loss.

The Signaling System of cAMP 5

Many hormones use the adenylate cyclase–cAMP second messenger system to stimulate their target tissues. The hormone binds with the receptor that is coupled to a G protein. The G protein stimulates the membrane-bound adenylate cyclase enzyme. Activated adenylate cyclase catalyzes the conversion of adenosine triphosphate (ATP) into cAMP in the cytoplasm.

The cAMP then activates an enzyme called cAMP-dependent protein kinase. This enzyme phosphorylates specific proteins in the cell, triggering a cascade of biochemical reactions that ultimately lead to the cell's response to the hormone.

exogenous gland with two examples 5

An exogenous gland is a gland that produces hormones or other substances that are released into the bloodstream or outside the body. Exogenous means caused by something outside the body. Examples of exogenous glands are:

The adrenal glands, which produce cortisol and other hormones that help regulate metabolism, blood pressure, and stress response. Exogenous adrenal insufficiency is a condition of low levels of adrenal hormones caused by factors such as long-term use of glucocorticoid medicines or removal of the adrenal glands.
The thyroid gland, which produces thyroxine and triiodothyronine, hormones that control the rate of energy use and growth in the body. Exogenous thyrotoxicosis is a condition of high levels of thyroid hormones caused by factors such as ingestion of thyroid hormone pills or exposure to iodine-containing substances.

Which parts that are organ of Corti?

The cochlear duct contains the sensory organ of hearing, the Organ of Corti that contains the hair
cells and transduces auditory stimuli into sensory signals.
It is the organ due to which human ear can detect and distinguish among different sound frequencies.
It is composed of:

- Hair cells
- Support cells
- A specialized basilar membrane
- Auditory nerve endings
- Tectorial membrane

physiology def

Animal Physiology deals with the study of functions of the tissues, organs and organ systems of
animals.
The ultimate goal of the study of physiology is to understand the mechanisms that operate in animals at all levels, in physical and chemical terms.

Effects of neuromodulator 5

The phenomenon of neuromodulation plays critical roles in the development of complex behavioral patterns e.g. happiness, revenge, reward, greed and exploration. Neuromodulation is
also involved in the processes of thinking, cognition, planning, learning and memory.
Many behavioral problems are also based on neuromodulation e.g. mood swings, sleep disturbances, feelings of stress, anxiety, depression and anger.

features of smooth muscles 3

These are the least specialized muscle fibers that have myosin similar to the form found in contractile nonmuscle cells.

➢ Smooth muscle fibers lack the characteristic striations produced by the organized

groups of actin and myosin filaments in sarcomeres.

➢ Smooth muscle of vertebrates is under autonomic and hormonal control and is not "voluntary".

➢ Smooth muscle cells contract and relax far more slowly than striated muscle fibers and generally are capable of more sustained contractions.

two examples of interoceptive receptors 3

Receptors of vestibular system that monitor the orientation of the body. They are present in the semicircular canal in inner ear.

• Thermoreceptors that keep track of thermal state of the body. They are found scattered on the skin.

• Chemoreceptors that keep track of chemical state of the body. Various types of these receptors are distributed in specific parts of the body.

• Proprioceptors that monitor the position of muscles and joints and are located in the movable joints.

• Nociceptors perceive the sensation of pain and are distributed evenly in the skin and deeper body parts.

• Pacinian corpuscles that are situated deep in the body and receive pressure stimulus. In the limbs, they receive vibrations.

How many groups of secondary messenger3

Second Messenger Types

The second messengers involved in signal transduction fall into three distinct groups:

• Cyclic nucleotide monophosphates e.g. cAMP (cyclic adenosine monophosphate) and cGMP (cyclic guanosine monophosphate).

• Inositol phospholipids e.g. inositol trisphosphate (InsP3) and diacylglycerol (DAG)

• Ca2+ ions and associated calmodulin

Properties of hormones 5

• Hormones act only on specific target cells/tissues

Although hormone molecules come into contact with all the tissues in the body during transport through blood, only the cells that contain receptors specific for a particular hormone are affected by it, and generate response.

• Hormonal action depend on the type of receptor

The action of a hormone depends on the nature of enzyme cascade linked to the hormone's receptor, as well as the effector molecules expressed in a particular tissue.

Write about Tetanus.

For this reason, a hormone can act on two or more different types of tissues and generate different types of responses.

For example Antidiuretic hormone stimulates water reabsorption in the nephron of kidney while the same hormone produces vasoconstriction and increases blood pressure by acting on the walls of arteries. The receptors in kidney nephron tubule and arterial receptors are different, causing the same hormone to generate different responses.

• Hormones are active at very low concentration

The amount of hormone produced by an endocrine gland is small. It is further diluted in the blood and interstitial fluid. The available concentration for the target cell lies between 10-8 to 10-12 M and the hormone is still effective at this very low concentration.
The high sensitivity of hormonal signaling is due to the high affinity of target cell receptors for hormones.
• Hormonal effects are amplified
Binding of a hormone molecule to its receptor leads to a cascade of enzymatic steps that amplify the effect; thus just a few hormone molecules can influence thousands or millions of molecular reactions within a cell

Name 3 glandular cells.
Three examples of glandular cells are:

Goblet cells: These are single-celled glands that secrete mucus to lubricate and protect the epithelial surfaces of the respiratory and digestive tracts.
Acinar cells: These are groups of cells that form spherical sacs called acini and secrete enzymes and other substances into ducts. Examples of acinar cells are the pancreatic acinar cells that produce digestive enzymes and the salivary acinar cells that produce saliva.
Follicular cells: These are groups of cells that form spherical structures called follicles and secrete hormones into the bloodstream. Examples of follicular cells are the thyroid follicular cells that produce thyroxine and triiodothyronine, and the ovarian follicular cells that produce estrogen.

range receptor classification and two examples marks5
Range receptor types encompass various sensory receptors that specialize in detecting specific types or ranges of stimuli. Two examples are:

Chemoreceptors: These receptors detect chemical stimuli such as taste and smell. They can detect the presence of specific molecules or changes in chemical concentrations. For instance, taste receptors on the tongue and olfactory receptors in the nasal cavity are examples of chemoreceptors.

Photoreceptors: These receptors are sensitive to light stimuli and are essential for vision. They can detect different wavelengths of light and are found in the retina of the eye. Rods and cones in the retina are examples of photoreceptors.

MCQs

1- synaptic cleft have diameter of… **20 nm**

2-what is not the function of Ca+ ion… **inhibit muscle contraction.**

3- the concentration of _ ion is **10 -20-fold** higher in inner side….

4-which is not the inhibitory harmone….

The concentration of K^+ is about 10-20 times higher inside cells than outside, while the opposite is true for Na^+.

The pituitary gland, also known as hypophysis is a small gland, about 1 centimeter in diameter and 0.5 to 1 gram in weight.

Mcqs

- physiology is the....animal..study of functions of tissues organs and organ system
- equation indicate blood flow.....v=Q/A
- cardiac output of resting person is.. **5L/M**
- Myocyte are.....**cardiac muscle**
- interstitial tissues constitute of adult tissue.....20%
- intestine is a target tissue of .....**PTH**
- human retina has....cones.....**6 million**

the reference value of resting membrane potential in neurons is taken as -70mV, when no impulse is being conducted.

- hair cells have resting potential.....**-60mv**
- the knob of dendrites have.....olfactory cilia....**4_ 25**
- which is not the part of cell body....**myllin sheath**
- Messiein carpusule are linked to.....**sense of touch**
- ACTH is a peptide hormone consist of....chain of amino acids....**39**
- ATPase is include.....**All**
- .....diffuse passively with the help of protein are..... **facilitated diffusion**

1.Adenohypophysis produce... **hormones**

.Human Sa node ___ serves as the **pacemaker** of the heart, generating **electrical signals** that initiate the heartbeat.

3.Adrenal cortex not produce….**Growth hormone**; it is primarily produced by the anterior pituitary gland.

3.Which is wrong about CA channel (**Larger than potassium (K+) channels)**

(Rise due to PA) **wrong**

Smaller then k+ **wrong**

4.Receptor for thyroid hormone

Cytoplasm

Neucleus

Both

Cell surface

5.Ca+ flow out of lumen Sr increase free ca+ in myoplasm

10-6

10-8